Tips and Tricks

and some Notes

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CHAPTER 1

THIS CHAPTER IS A TEST CHAPTER, TO SEE IF EVERYTHING IS IN ORDER, IN CASE THINGS BLOW UP

For example, here we try to apply the thmp, thm, lemp, def, and see if they work. also test for other subtleties

Definition 0.1: Compact set

A set K in a topological space $X, \mathbb{T}_X$ is compact if for every open cover $\mathbb{U}$ of K, there exists a finite subcover $\mathbb{U}'$ of $\mathbb{U}$ that also covers K.

Example : The closed interval $[0, 1]$ ***in the real numbers*** $\mathbb{R}$ ***with the standard topology is a compact set. This is because any open cover of*** $[0, 1]$ ***can be reduced to a finite subcover that still covers the entire interval.***